the closure of the tricuspid and bicuspid valves at the beginning of ventricular systole whereas second heart sound dub is associated with the closure of the semilunar valves at the end of ventricular systole. These sounds are of clinical diagnostic significance. An increased heart rate is called **tachycardia** and decreased heart rate is called **bradycardia**.

7.4.2 Cardiac Cycle

The events that occur at the beginning of heart beat and lasts until the beginning of next beat is called cardiac cycle. It lasts for 0.8 seconds. The series of events that takes place in a **cardiac cycle** are as follows;

PHASE 1**: Ventricular diastole**- The pressure in the auricles increases than that of the ventricular pressure. AV valves are open while the semi lunar valves are closed. Blood flows from the auricles into the ventricles passively.

PHASE 2: **Atrial systole** - The atria contracts while the ventricles are still relaxed. The contraction of the auricles pushes maximum volume of blood to the ventricles until they reach the end diastolic volume (EDV). EDV is related to the length of the cardiac muscle fibre. More the muscle is stretched, greater the EDV and the stroke volume.

PHASE 3: V**entricular systole** (isovolumetric contraction) - The ventricular contraction forces the AV valves to close and increases the pressure inside the ventricles. There is no change in the size of the muscle fibre length and ventricular chamber volume (isovolumetric contraction).

PHASE 4: **Ventricular systole** (ventricular ejection) - Increased ventricular pressure forces the semilunar valves to open and blood is ejected out of the ventricles through the aorta and pulmonary artery without backflow of blood. This point is the end of systolic volume (ESV).

PHASE 5: (**Ventricular diastole)** -The ventricles begins to relax, pressure in the arteries exceeds ventricular pressure, resulting in the closure of the semilunar valves. The heart returns to phase 1 of the cardiac cycle.

7.4.3 Cardiac Output

The amount of blood pumped out by each ventricle per minute is called **cardiac output(CO)**. It is a product of **heart rate** (HR) and **stroke volume** (SV). Heart rate or pulse is the number of beats per minute. Pulse pressure = systolic pressure – diastolic pressure. Stroke volume (SV) is the volume of blood pumped out by one ventricle with each beat. SV depends on ventricular contraction.

$$CO = HR \times SV$$

SV represents the difference between EDV (amount of blood that collects in a ventricle during diastole) and ESV (volume of blood remaining in the ventricle after contraction). SV = EDV - ESV.

According to Frank – Starling law of the heart, the critical factor controlling SV is the degree to which the cardiac muscle cells are stretched just before they contract. The most important factor in stretching cardiac muscle is the amount of blood returning to the heart and distending its ventricles, venous return. During vigorous exercise, SV may double as a result of venous return. Heart's pumping action normally maintains a balance between cardiac output and venous return. Because the heart is a double pump, each side can fail

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